# A Comprehensive Measure of Well-Being

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## Project Title

**A Comprehensive Measure of Well-Being: Human Development Index (HDI) Prediction System Using Machine Learning**

## Abstract

The Human Development Index (HDI) Prediction System is a machine learning-based web application developed to predict the Human Development Index of a country using important socio-economic indicators such as life expectancy, education level, and Gross National Income (GNI) per capita. The project uses a Linear Regression algorithm to analyze the relationship between these indicators and HDI. The trained model is deployed using the Flask web framework, allowing users to enter input values through a web interface and receive HDI predictions instantly. This project demonstrates the application of machine learning techniques in solving real-world problems related to human development and policy analysis.

## Problem Statement

The Human Development Index is widely used to measure the overall development of a country by considering health, education, and income rather than economic growth alone. Predicting HDI manually is difficult because multiple indicators influence it simultaneously. This project aims to develop a machine learning model capable of accurately predicting HDI using socio-economic features. The application helps researchers, policymakers, and students analyze development trends and identify areas requiring improvement.

## Scenarios

**Scenario 1:** Predicting Very High Human Development

A user selects a country with high life expectancy, strong mean years of schooling, expected years of schooling, and a high GNI per capita. Based on these indicators, the model predicts a Very High HDI score, classifying the country among the most developed nations. This helps users understand how strong performance across multiple development dimensions contributes to overall human development.

**Scenario 2:** Identifying Development Gaps in Emerging Economies

A policymaker inputs mid-range values representing a developing nation, including moderate life expectancy, average educational attainment, and a moderate income level. The model returns a Medium HDI score, providing insights into areas where improvements in healthcare, education, or income generation could significantly enhance human development outcomes.

**Scenario 3:** Assessing Countries Requiring Development Intervention

A researcher evaluates a country with relatively low life expectancy, limited educational opportunities, and low GNI per capita. The model predicts a Low HDI score and highlights the country's developmental challenges. Such predictions can support governments and development organizations in prioritizing investments and policy interventions.

## Objectives

- Predict the Human Development Index using Machine Learning.
- Analyze the relationship between HDI and socio-economic indicators.
- Build a Flask web application for user interaction.
- Deploy the trained machine learning model for real-time predictions.

## Features

- Human Development Index Prediction
- Machine Learning-based Prediction
- User-friendly Flask Web Application
- Interactive HTML Interface
- Trained Linear Regression Model
- Fast and Accurate Prediction
- Data Visualization using Matplotlib and Seaborn

## PREREQUISITES:

The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.

Anaconda Navigator: A desktop graphical user interface used to launch applications and manage Python packages and environments for data science. https://www.anaconda.com/download

PyCharm: An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools. https://www.jetbrains.com/pycharm/

NumPy: Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. https://numpy.org/doc/stable/

Pandas: Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series. https://pandas.pydata.org/docs/

Scikit-learn: Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. https://scikit-learn.org/stable

Matplotlib: Comprehensive library for creating static, animated, and interactive visualizations in Python. https://matplotlib.org/stable/

Seaborn: Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics. https://seaborn.pydata.org/

Flask: A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. https://flask.palletsprojects.com/

## Technologies Used

Programming Language

- Python

Framework

- Flask

Machine Learning

- Scikit-learn

Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Pickle

Frontend

- HTML
- CSS

IDE

- Visual Studio Code
- Jupyter Notebook / Anaconda

## Dataset Description

The dataset used in this project is collected from Kaggle and contains Human Development indicators for different countries.

Dataset includes:

- Country
- HDI Score
- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income Per Capita
- Other Human Development Indicators

## Project Flow

1. Environment Setup
2. Install Required Libraries
3. Import Required Packages
4. Load Dataset
5. Data Visualization
6. Data Preprocessing
7. Handle Missing Values
8. Split Dataset into Train/Test
9. Train Linear Regression Model
10. Evaluate Model
11. Save Model using Pickle
12. Develop Flask Application
13. Predict HDI through Web Interface

## Usage Instructions

1. Launch the Flask application.
2. Open the application in your browser.
3. Enter the required Human Development indicators.
4. Click the **Predict** button.
5. View the predicted HDI score

## Environment Setup and Package Installation

- Install Python
- Install Flask
- Install Scikit-learn
- Install NumPy
- Install Pandas
- Install Matplotlib
- Install Seaborn

# Folder Structure

```
> OPEN EDITORS
v HUMANINDEX
  v ML - 0027 - HUMAN DEVELOPMENT INDEX-...
    v Dataset
        HDI.csv
    v Flask
      v static
          correlation_heatmap.png
          distribution_plots.png
          scatter_gni_hdi.png
          style.css
      v templates
          index.html
          predict.html
          result.html
        app.py
        HDI.pkl
    v Training
        HumDevIndex.ipynb
  > venv
    requirements.txt
```

---

# Import Required Libraries

Import:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle
- Flask

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import pickle
```

## Dataset Download and Understanding

- Download dataset from Kaggle
- Load CSV dataset
- Analyze features
- Display dataset information
- Perform Exploratory Data Analysis

```
df = pd.read_csv('../Dataset/HDI.csv')
df.head()
```

---

## Data Preprocessing

- Select input features
- Handle missing values
- Label Encoding (if required)
- Feature Selection
- Prepare training dataset

```
plt.figure(figsize=(8,6))
sns.heatmap(df.drop(columns=['Country']).corr(), annot=True, cmap='coolwarm')
plt.show()
```

---

## Train-Test Split

Split dataset into:

- Training Data
- Testing Data

```
features = ['Life Expectancy', 'Mean Years of Schooling', 'Expected Years of Schooling', 'GNI Per Capita']
for feature in features:
    df[feature] = df[feature].fillna(df[feature].mean())
X = df[features]
y = df['HDI Score']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

## Model Training

- Train Linear Regression Model
- Generate Predictions
- Evaluate Model
- Calculate $R^2$ Score

```python
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f'R-Squared: {r2_score(y_test, y_pred)}')
with open('../Flask/HDI.pkl', 'wb') as f:
    pickle.dump(model, f)
```

## Save Model

Save model using Pickle.

```python
with open('../Flask/HDI.pkl', 'wb') as f:
    pickle.dump(model, f)
```

---

## Flask Web Application

- Create Flask App
- Create HTML Templates
- Load Saved Model
- Accept User Inputs
- Predict HDI
- Display Results

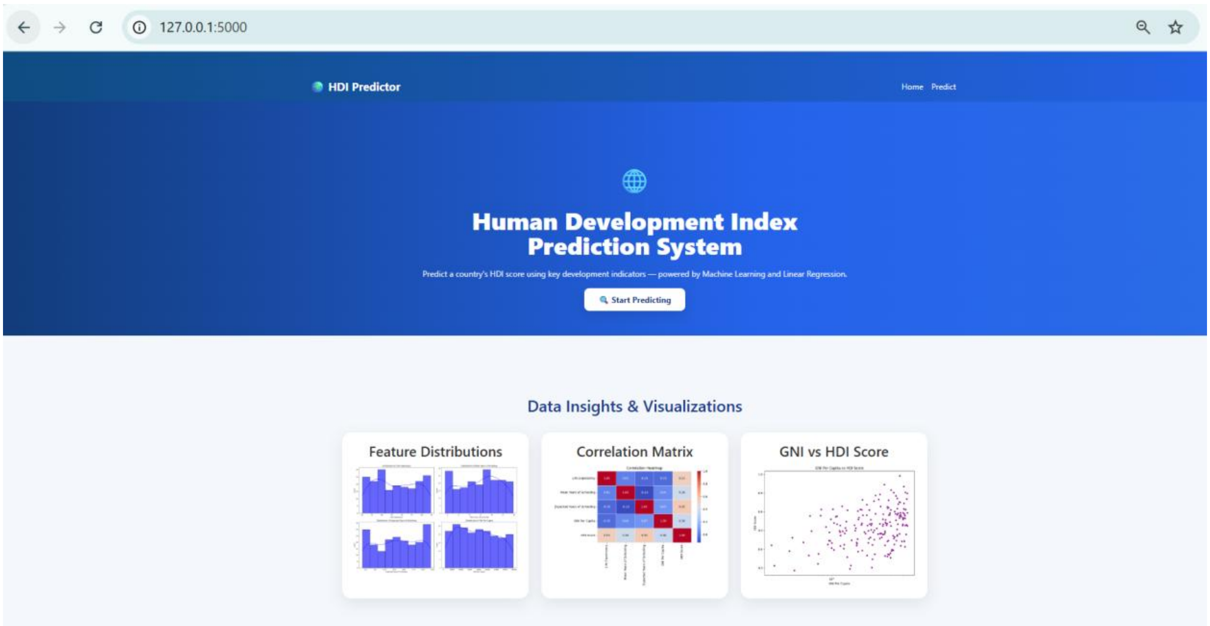
Index.html

```html
ML - 0027 - HUMAN DEVELOPMENT INDEX-202605 > ML - 0027 - Human Development Index > Flask > templates > index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Human Development Index (HDI) Predictor</title>
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
        <nav class="navbar navbar-expand-lg navbar-dark hdi-navbar">
            <div class="container">
                <a class="navbar-brand fw-bold" href="/">🌍 HDI Predictor</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navMenu">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="collapse navbar-collapse" id="navMenu">
                    <ul class="navbar-nav ms-auto">
                        <li class="nav-item"><a class="nav-link" href="/">Home</a></li>
                        <li class="nav-item"><a class="nav-link" href="/predict">Predict</a></li>
                    </ul>
                </div>
            </div>
        </nav>

        <section class="hero text-center">
            <div class="container">
                <div class="hero-icon">🌐</div>
                <h1 class="hero-title">Human Development Index<br>Prediction System</h1>
                <p class="hero-sub">Predict a country's HDI score using key development indicators — powered by Machine Learning and Linear Regression.</p>
                <a href="/predict" class="btn-hero">🔍 Start Predicting</a>
            </div>
        </section>

        <main class="section-pad">
            <div class="container">
```

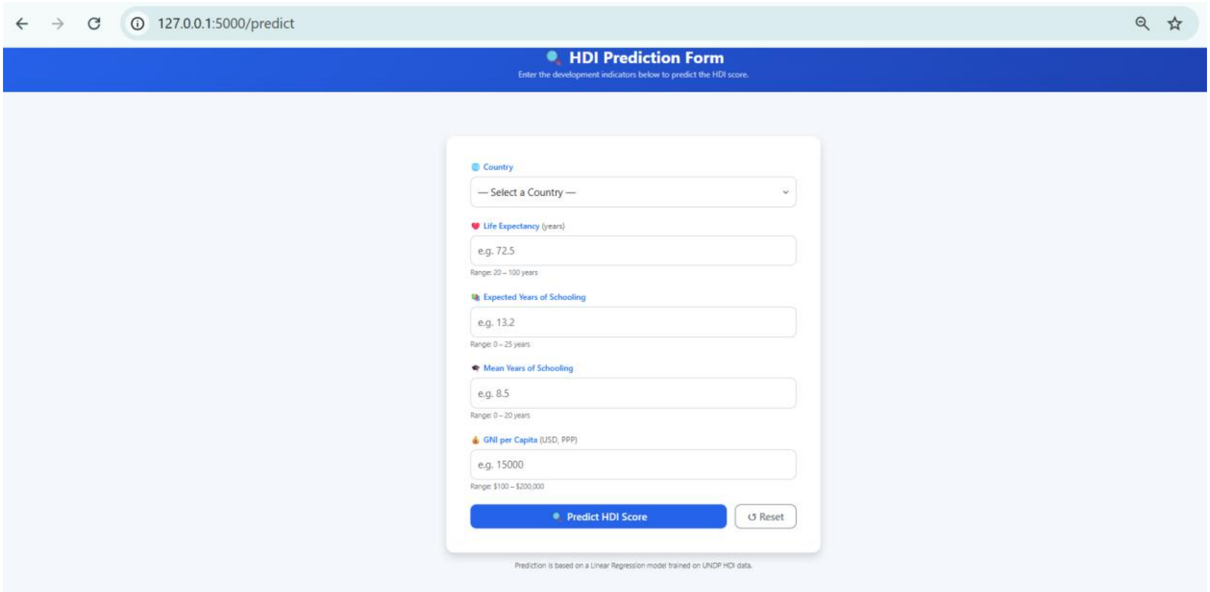
Predict.html

```
M Welcome HUMAN DEVELOPMENT INDEX-202605 > ML - 0027 - Human Development Index > Flask > templates > <> predict.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Predict HDI – HDI Prediction System</title>
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}" />
</head>
<body>

<!-- ── Navbar ──────────────────────────────────────────── -->
<nav class="navbar navbar-expand-lg navbar-dark hdi-navbar">
  <div class="container">
    <a class="navbar-brand fw-bold" href="/">🌍 HDI Predictor</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link" href="/">Home</a></li>
        <li class="nav-item"><a class="nav-link active" href="/predict">Predict</a></li>
      </ul>
    </div>
  </div>
</nav>

<!-- ── Page Header ─────────────────────────────────────── -->
<div class="page-header text-white text-center py-4">
  <div class="container">
    <h2 class="fw-bold mb-1">🔍 HDI Prediction Form</h2>
    <p class="opacity-75 mb-0">Enter the development indicators below to predict the HDI score.</p>
  </div>
</div>

<!-- ── Form ────────────────────────────────────────────── -->
<section class="py-5">
```

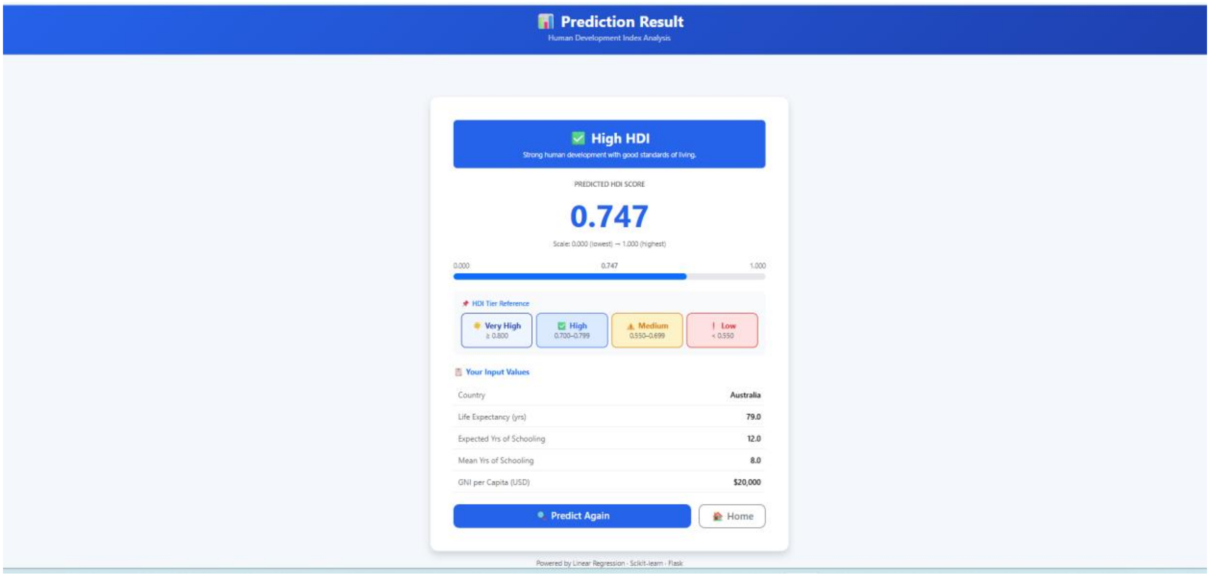
Home Page



Predicting Page




Output:



## Future Scope

- Improve prediction accuracy using advanced machine learning algorithms.
- Deploy the application on cloud platforms.
- Integrate real-time HDI datasets.
- Add interactive dashboards and data visualizations.
- Develop a mobile application version.
- Support prediction for multiple years.

## Conclusion

The Human Development Index Prediction System demonstrates how machine learning can assist in analyzing and predicting human development using socio-economic indicators. The project successfully integrates data analysis, machine learning, and Flask web development into a user-friendly application. It provides valuable insights for students, researchers, and policymakers while showcasing practical implementation of Artificial Intelligence in social and economic development.

Github link: https://github.com/Vinaymohan1234/A-Comprehensive-Measure-of-Well-Being

Demo Video Link: https://drive.google.com/file/d/1C6aRpMUzr5EPvoUu1_xWcZsN-oBqMnEV/view?usp=sharing